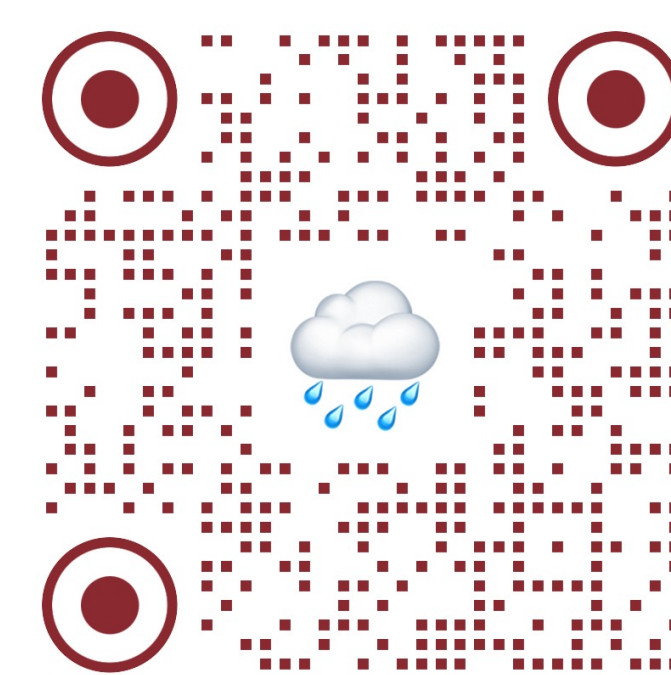


Is there a strategy switch cost when switching strategies within one task?

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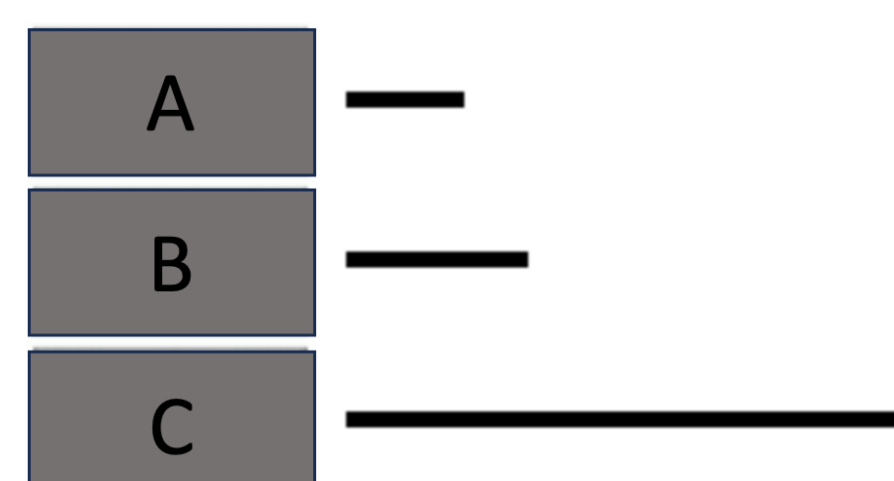
Introduction

- Switching between **tasks** often incurs a **task switch cost**: increased reaction time (RT) and/or reduced accuracy.
- But: **Does switching strategies within the same task** without changing the task itself lead to similar performance costs?
- We define **strategy switch cost** as increased reaction time and/or decreased accuracy in task performance when switching strategies.
- This study aims to investigate the presence of **strategy switch costs** and explore whether there are individual differences in strategy switch costs.

Methods (N =96)

Building Sticks Task (BST; Lovett & Schunn, 1999)

• Undershoot



• Overshoot



Procedure

Pure 1

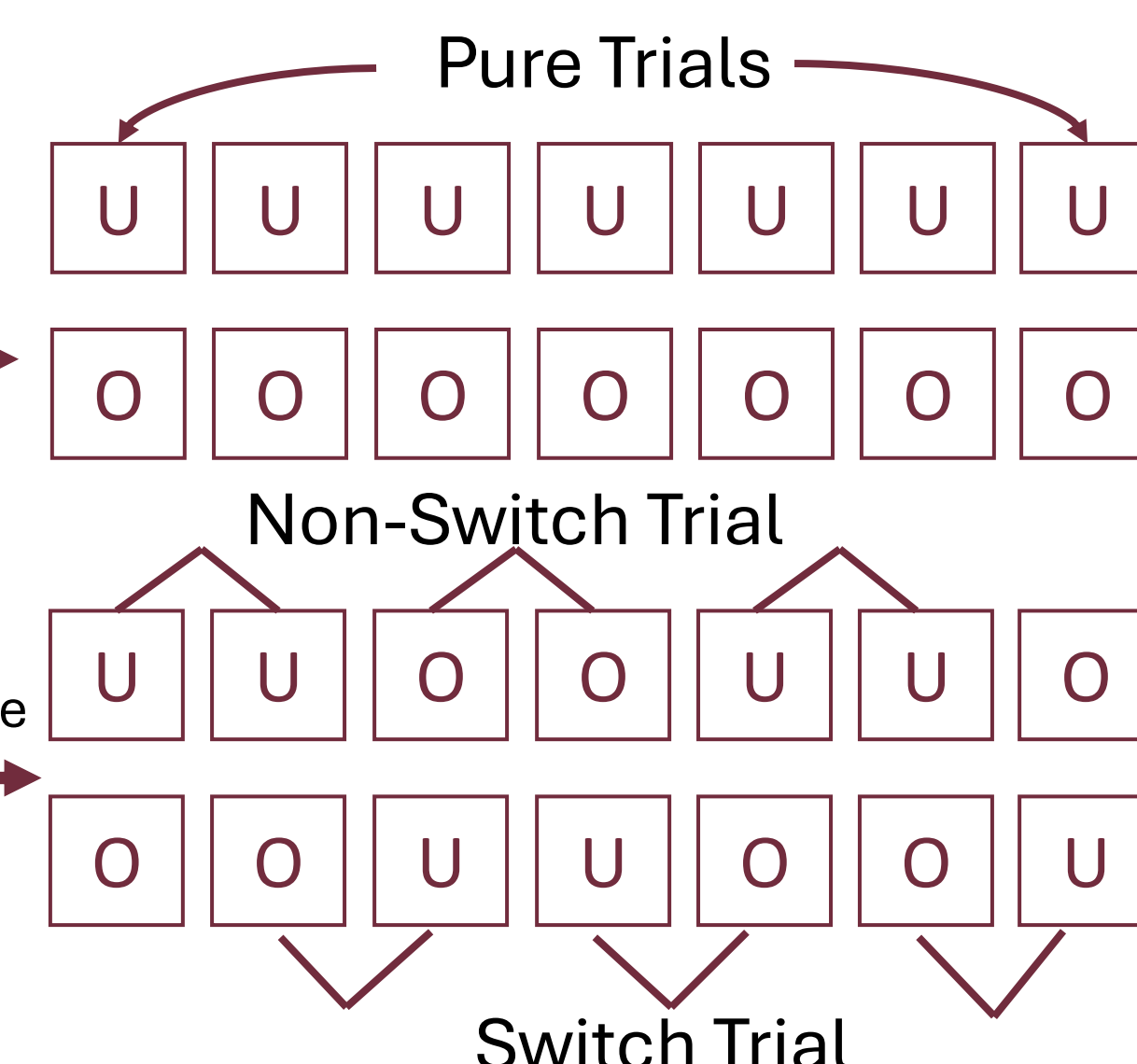
Pure 2

Mixed 3

Mixed 4

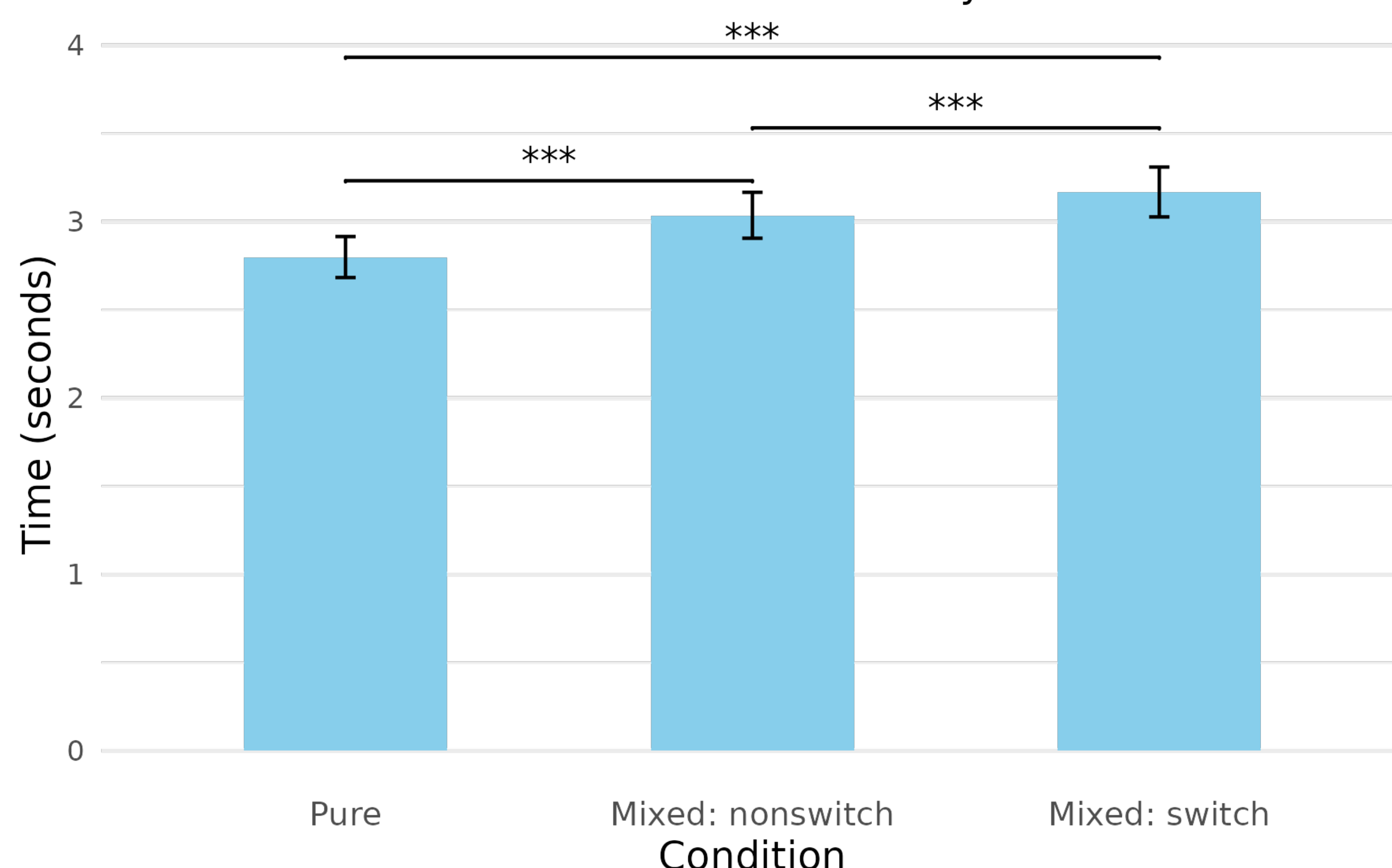
Random Sequence

Alternating Sequence

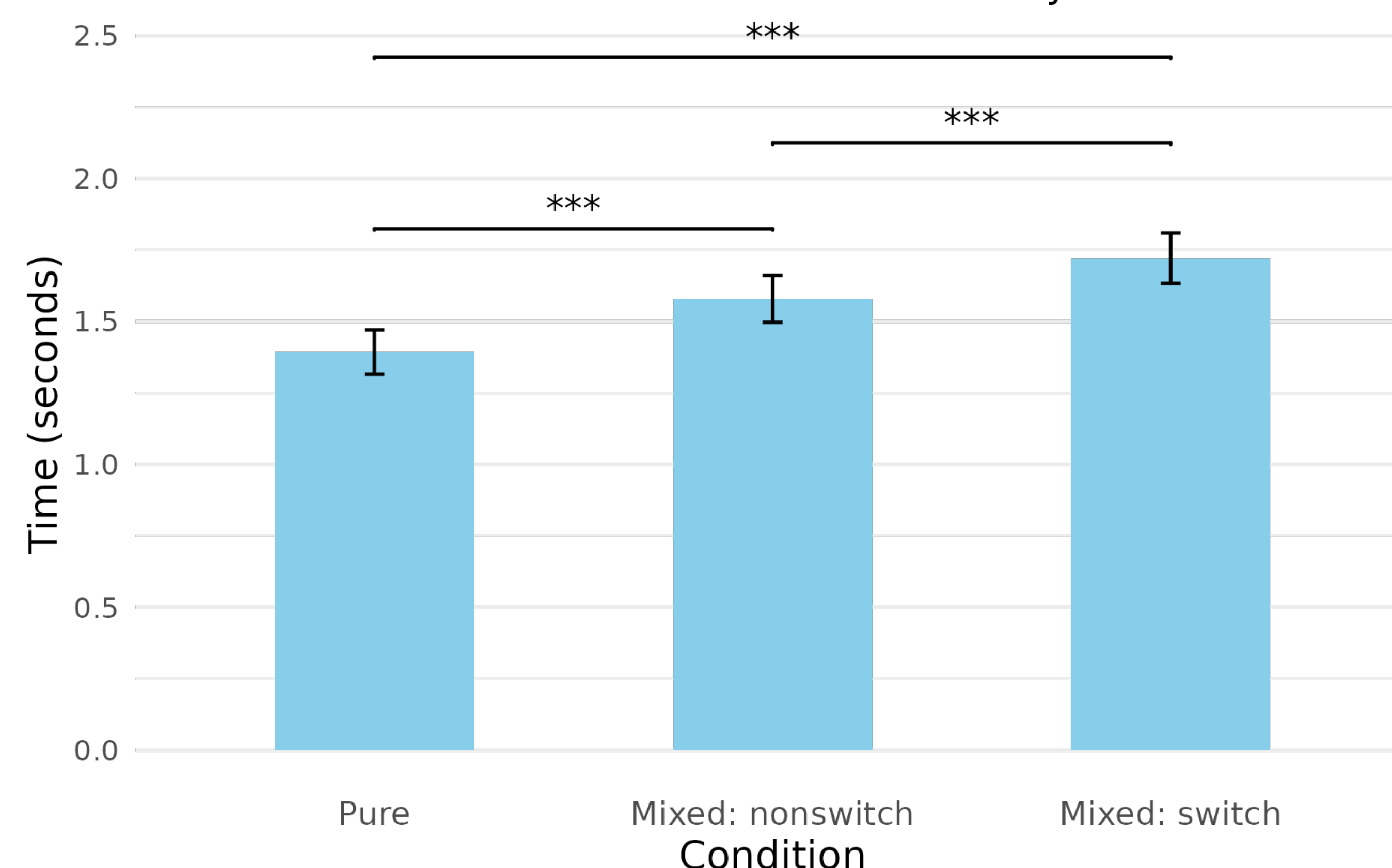


Results and Discussion

Estimated Mean Solution Time by Condition



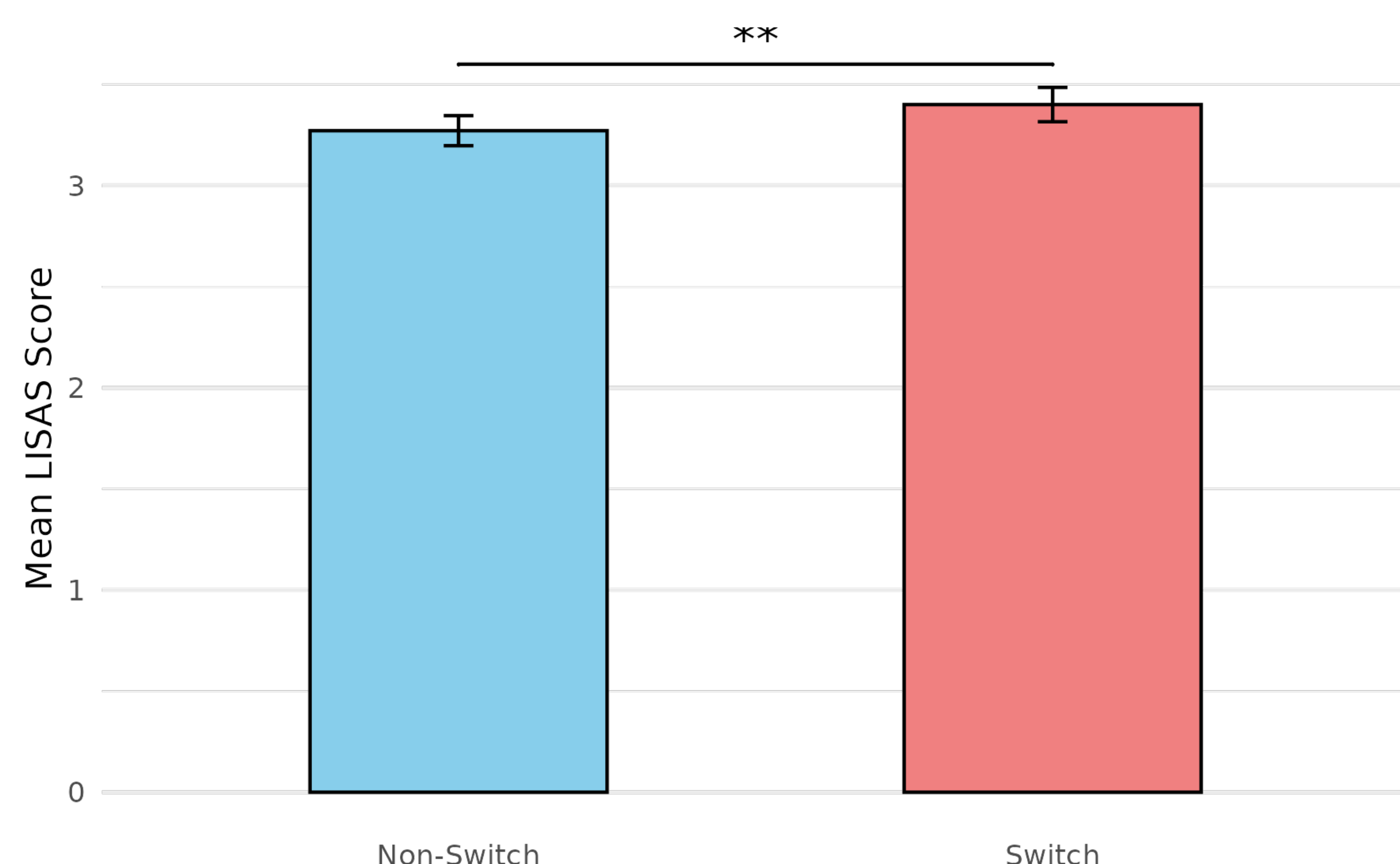
Estimated Mean Initial Reaction Time by Condition



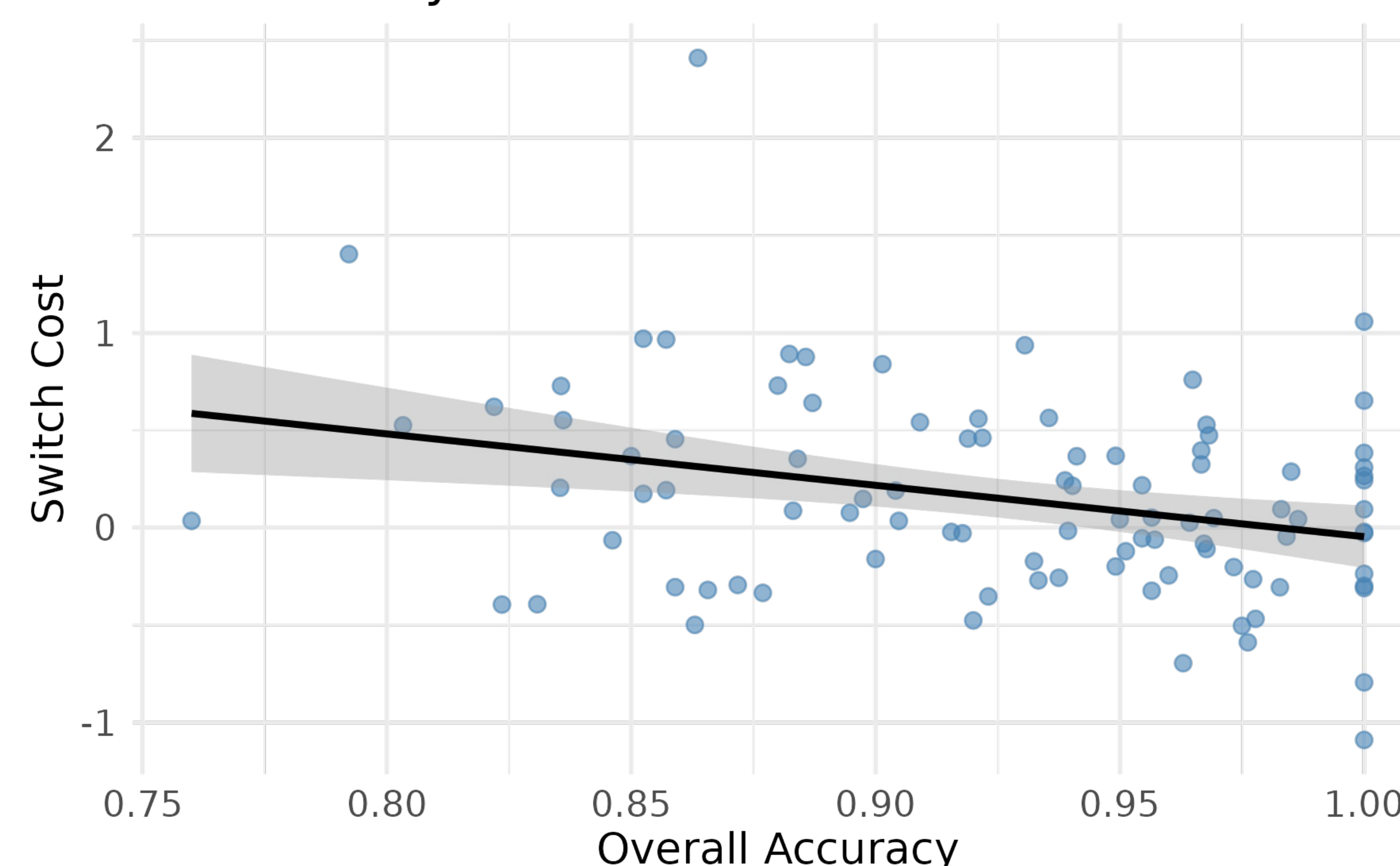
- Switch trials had significantly longer solution times than non-switch trials, $b = 0.043$, $SE = 0.013$, $t = 3.38$, $p < .001$.
- Both switch and non-switch trials were significantly slower than pure trials ($ps < .001$), indicating clear strategy switch costs.

- Switch trials had significantly longer initial reaction times than non-switch trials, $b = 0.142$, $SE = 0.033$, $t = 4.35$, $p < .001$.
- Both switch and non-switch trials were significantly slower than pure trials ($ps < .001$).

Switch vs. Non-Switch Trials in Mixed Blocks



Accuracy and Switch Cost in Solution Time



- LISAS: linear integrated speed-accuracy score (Barulli et al, 2023).
- LISAS = (Mean Proportion Errors (PE) on switch trials x (Standard Deviation of RT / Standard Deviation of PE)) + Mean Switch Trial RT (for correct trials only)
- Switch trials had higher LISAS than non-switch trials in mixed blocks, indicating a strategy switch cost, $t(94) = 2.72$, $p = .008$.

- Switch cost = switch trials – non-switch trials
- Higher accuracy significantly predicted smaller switch costs in solution time, $b = -2.63$, $SE = 0.86$, $t(93) = -3.07$, $p = .003$, but not for initial response time, $p = .107$.

Acknowledgement

This work was supported by Office of Naval Research grant N00014-21-1-2617.

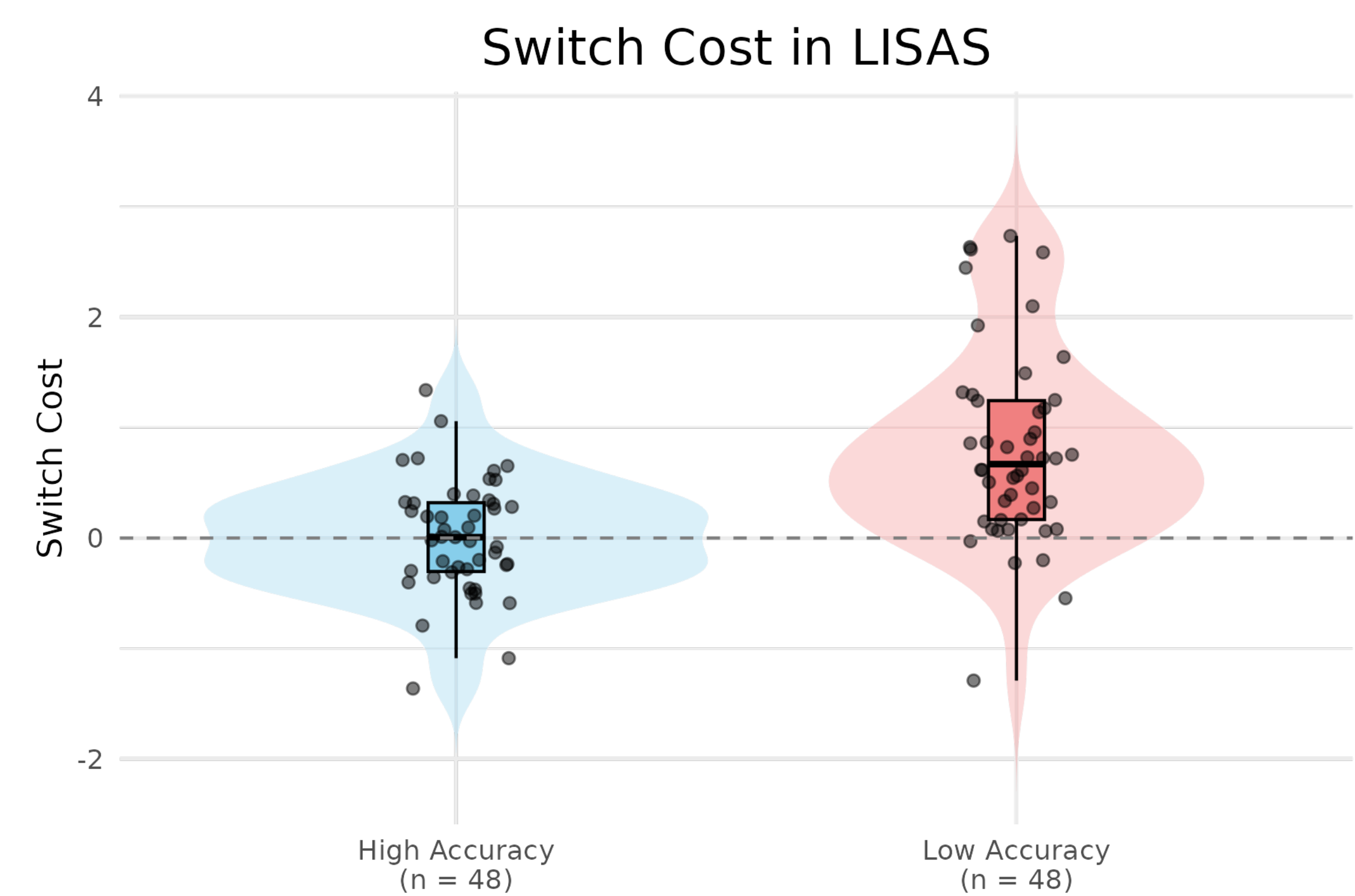
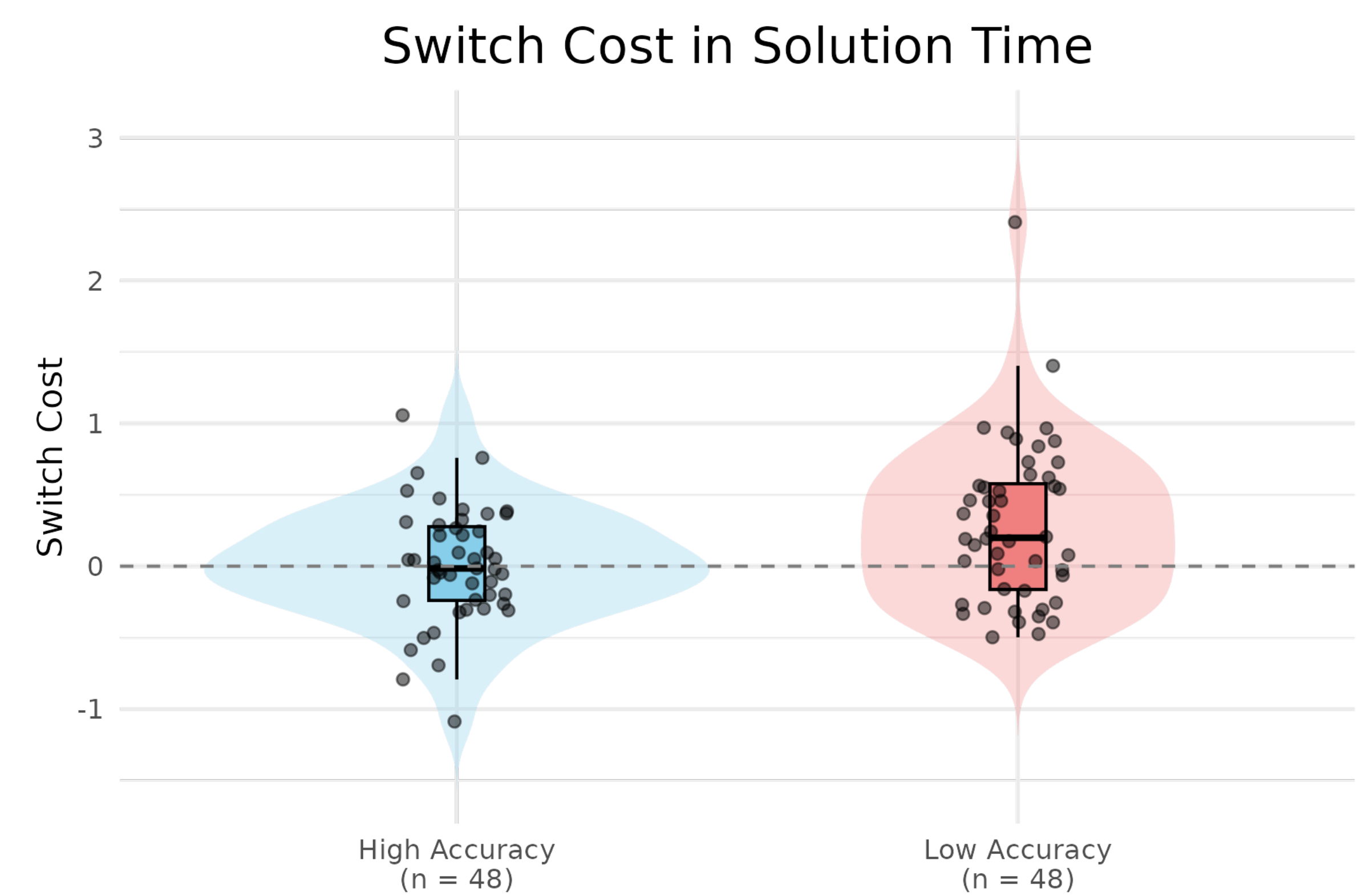
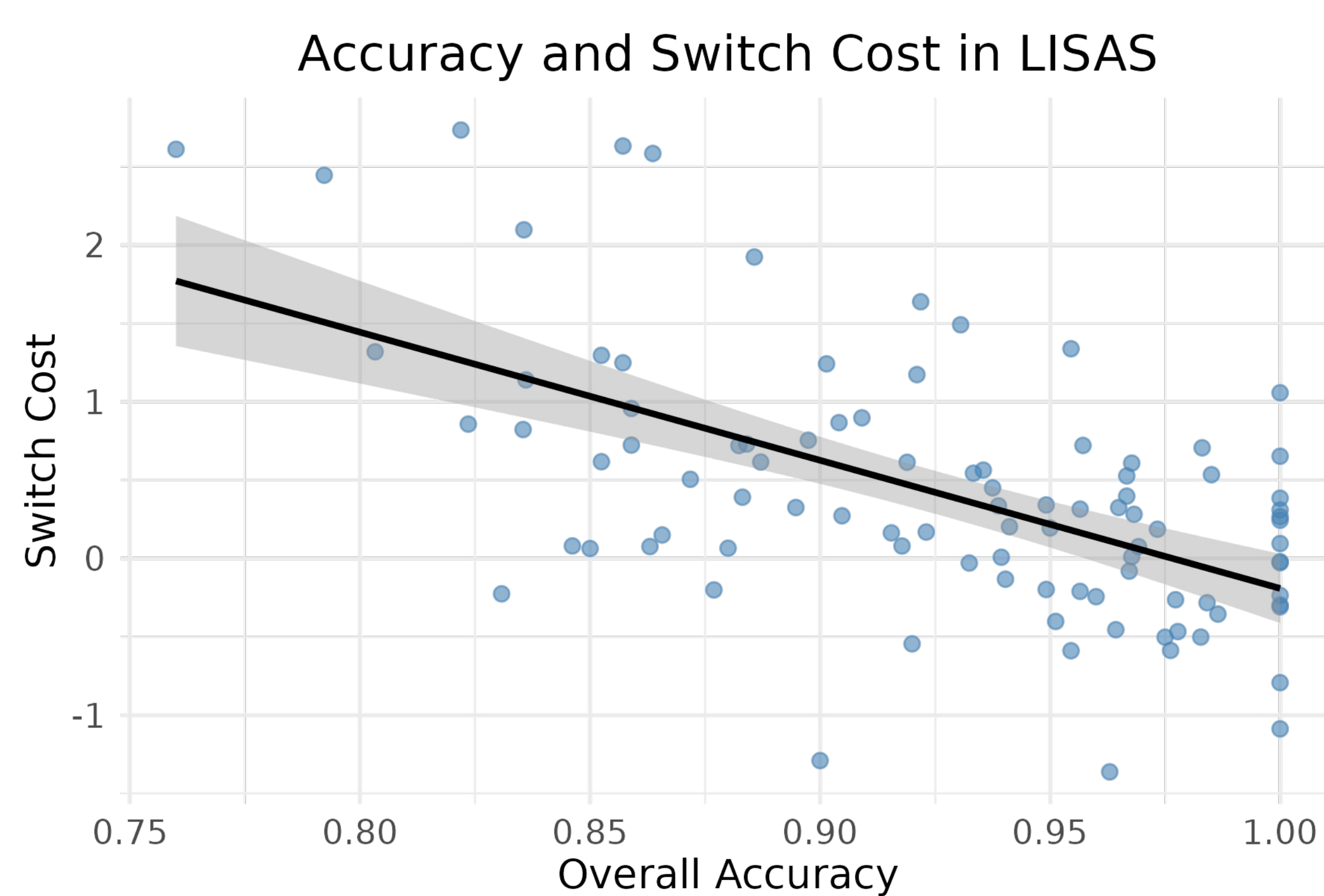
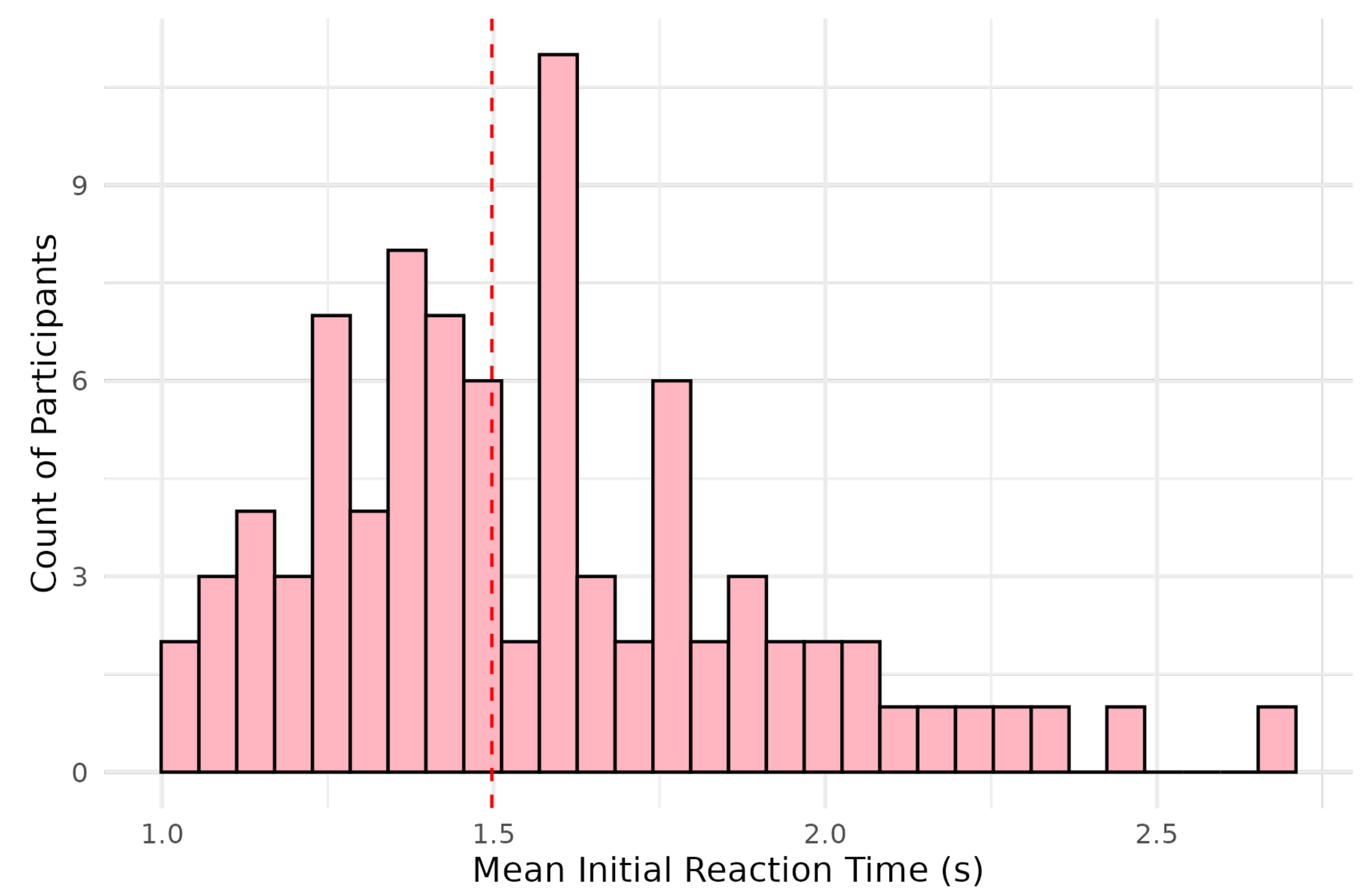
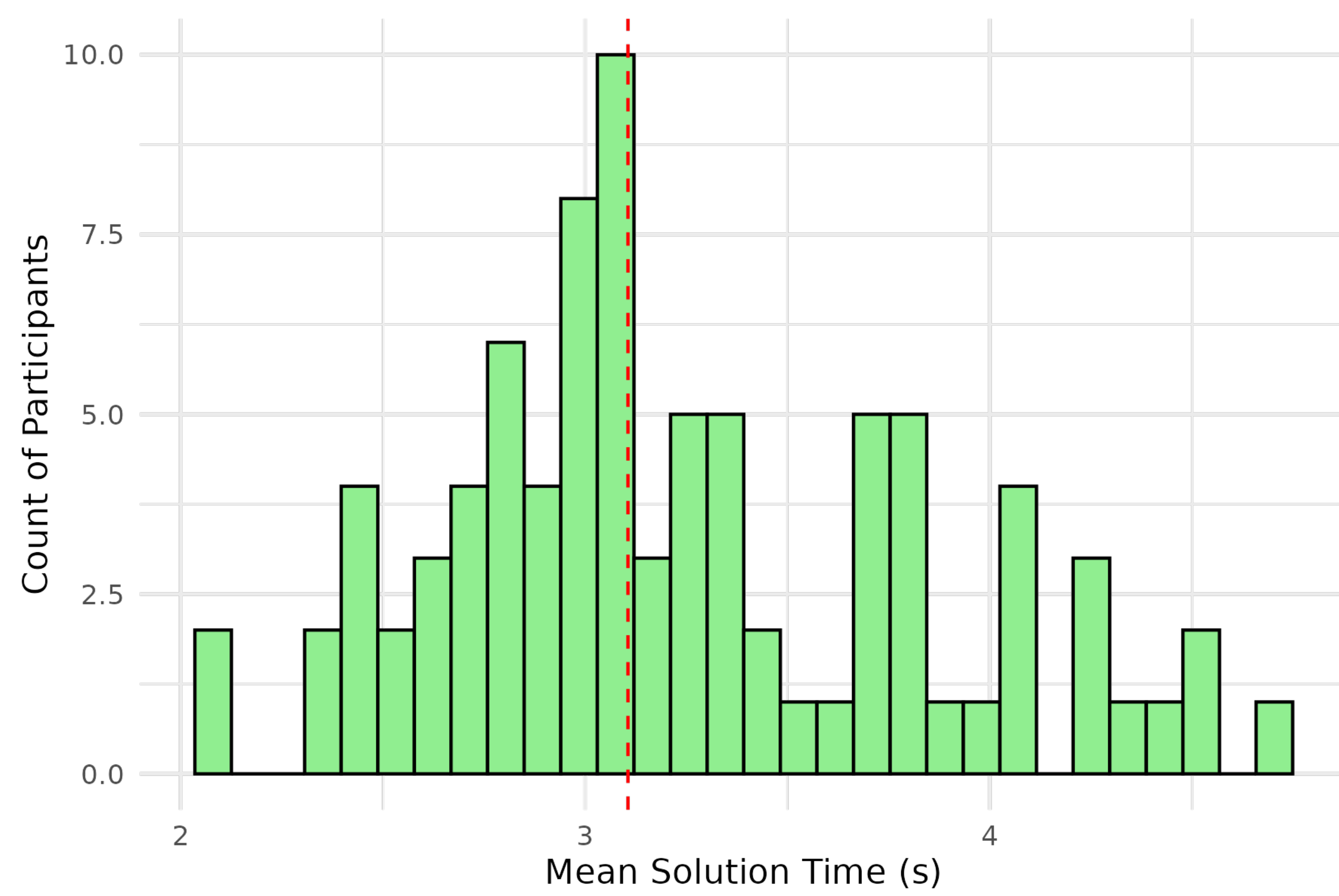
Key Takeaways:

- Strategy switch costs are real: switching strategies slows down performance. Strategy switching led to longer solution time and initial reaction time and higher LISAS.
- Higher accuracy buffers against switch costs in solution time but not in initial decision speed.

References & supplementary



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